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Reviewer activities: Army MI, MU, AT
Navy AS, SH, OS
Air Force 11, 99
DSA - ES

User activities: Army ME, AV, WC
Navy MC
Air Force

DD FORM 1 SEP 63 672

P.A Navy - EC

Other Cust
AF - 85
Army - EL

Procurement Specification
MIL-C-28748

International interest:

SUPERSEDES:

TITLE INSERT (INSULATOR), RECTANGULAR CONNECTOR, POLARIZED CENTER SCREW-LOCK, ELECTRICAL, 80 (SIZE 16 CRIMP REMOVABLE) PIN CONTACTS

MS 18163

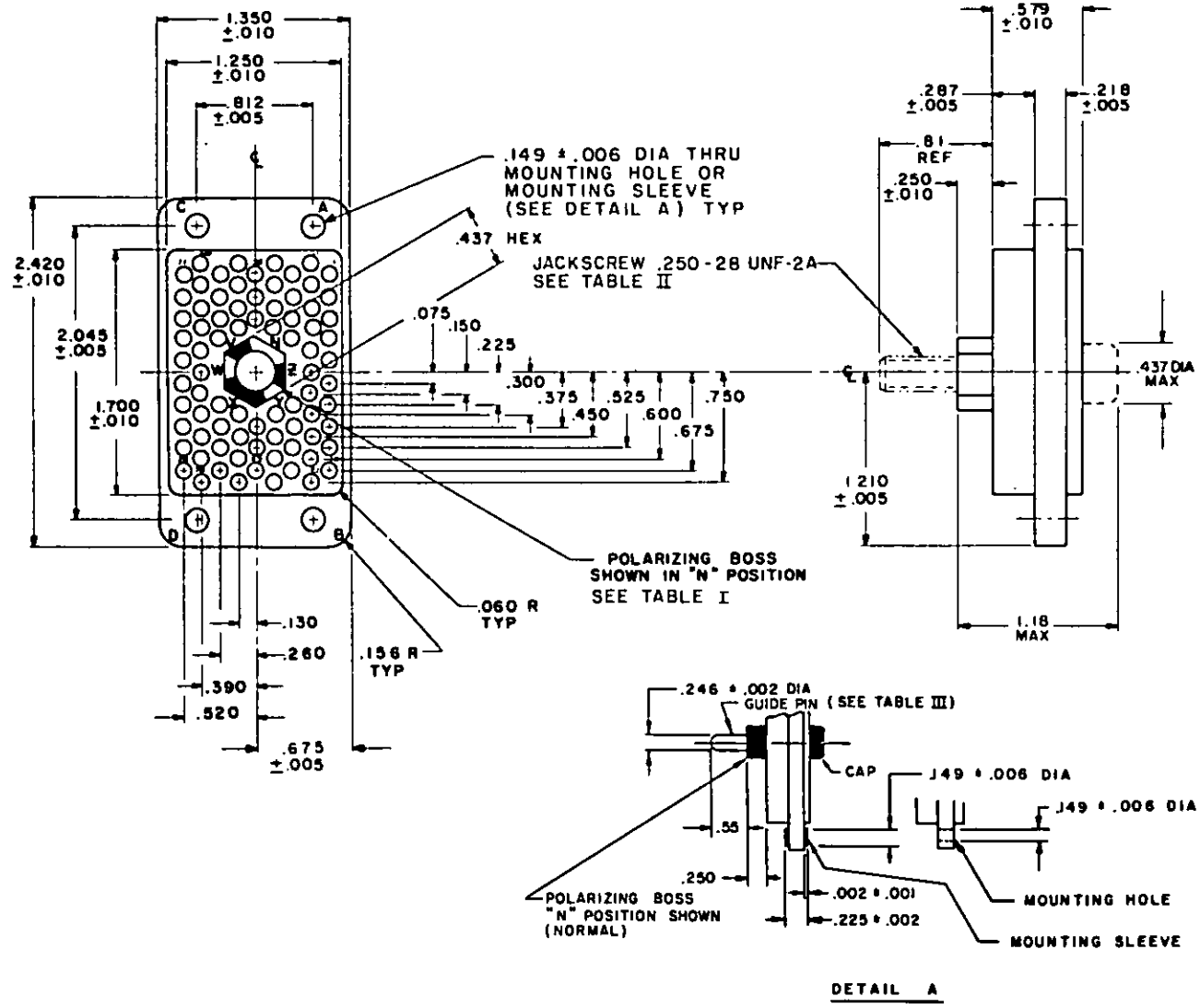
MILITARY STANDARD

SHEET 1 OF 3

5935-2004-13

Design standard for use with MIL-C-28748/9

(B) Entire standard revised



APPROVED 20 Dec 1965 REVISED (A) 20 May 1970 (B) 8 July 1976

FED. SUP CLASS 5935

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TYPE
SEE TABLE I

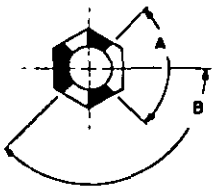


TABLE I		
TYPE	A	B
1	90°	135°
2	85°	137°30'

TABLE II		
MS PART NUMBER	POLARIZATION *	JACKSCREW
MS 18163-1N	N (TYPE 1)	SUPPLIED WITH CONNECTOR
MS 18163-1V	V (TYPE 1)	
MS 18163-1W	W (TYPE 1)	
MS 18163-1X	X (TYPE 2)	
MS 18163-1Y	Y (TYPE 2)	
MS 18163-1Z	Z (TYPE 2)	
MS 18163-2N	N (TYPE 1)	NOT SUPPLIED WITH CONNECTOR POLARIZING BOSS THREADED INTERNALLY .250-28 UNF-2B
MS 18163-2V	V (TYPE 1)	
MS 18163-2W	W (TYPE 1)	
MS 18163-2X	X (TYPE 2)	
MS 18163-2Y	Y (TYPE 2)	
MS 18163-2Z	Z (TYPE 2)	

TABLE III		
MS PART NUMBER	POLARIZATION *	GUIDE PIN
MS 18163-3N	N (TYPE 1)	SUPPLIED WITH CONNECTOR
MS 18163-3V	V (TYPE 1)	
MS 18163-3W	W (TYPE 1)	
MS 18163-3X	X (TYPE 2)	
MS 18163-3Y	Y (TYPE 2)	
MS 18163-3Z	Z (TYPE 2)	
MS 18163-4N	N (TYPE 1)	NOT SUPPLIED WITH CONNECTOR POLARIZING BOSS WITH .254 ± .002 I.D. TO ACCEPT GUIDE PIN
MS 18163-4V	V (TYPE 1)	
MS 18163-4W	W (TYPE 1)	
MS 18163-4X	X (TYPE 2)	
MS 18163-4Y	Y (TYPE 2)	
MS 18163-4Z	Z (TYPE 2)	

* LARGE OPENING IN POLARIZATION BOSS MUST BE ADJACENT TO POLARIZATION LETTER ON CONNECTOR BODY.

APPROVED 20 Dec 1965 REVISED (B) For changes see sheets 1, 2 and 3

User activities: Army - ME, AV, WC
Navy - MC
Air Force

Reviewer activities: Army - MI, MU, AT
Navy - AS, SH, OS
Air Force - 11, 80
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P.A Navy - EC	International interest:	TITLE INSERT (INSULATOR), RECTANGULAR CONNECTOR, POLARIZED CENTER SCREW-LOCK, ELECTRICAL, 80 (SIZE 16 CRIMP REMOVABLE) PIN CONTACTS	MILITARY STANDARD
Other Cust AF - 85 Army - EL			MS 18163
Procurement Specification MIL-C-28748		SUPERSEDES:	SHEET 2 OF 3

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INCHES	MM	INCHES	MM
.001	.03	.390	9.91
.002	.05	.437	11.10
.005	.13	.450	11.43
.006	.15	.520	13.21
.010	.25	.525	13.34
.060	1.52	.550	13.97
.075	1.91	.579	14.71
.130	3.30	.600	15.24
.149	3.79	.675	17.15
.150	3.81	.750	19.05
.156	3.96	.810	20.57
.218	5.54	.812	20.63
.225	5.72	1.180	29.97
.246	6.25	1.210	30.73
.250	6.35	1.250	31.75
.260	6.60	1.350	34.29
.287	7.29	1.700	43.18
.300	7.62	2.045	51.94
.375	9.53	2.420	61.47

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.015 (.38 mm).
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Tolerance between any two adjacent contact centers shall be ±.004 (.10 mm). Tolerance between any two contact centers, other than adjacent contacts, shall be ±.006 (.15 mm).
5. Contact hole details shall be in accordance with MIL-C-28748.
6. For mating connector see MS18164 (socket body).
7. Dimensions symmetrical about centerlines.
8. For design feature purposes, this standard takes precedence over procurement documents referenced herein.
9. Reference documents shall be of the issue in effect on date of invitation for bid.
10. Insert material shall be in accordance with MIL-C-28748.
11. Flatness, squareness and parallelism shall be within dimensional tolerances.
12. Center polarizing hardware may be rotated by removing retaining ring and spacer or removing cap (left hand thread) at rear of insert.
13. Maximum insertion and withdrawal force (connector assembly) shall be 75 pounds.
14. Maximum torque required at jackscrew to properly engage or disengage connector shall be 6 inch pounds.
15. Jackscrew wrench size 5/32 hex.

User activities: Army. ME, AV, WC
Navy MC
Air Force

Reviewer activities: Army MI, MU, AT
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P.A Navy - EC Other Cust AF - 85 Army - EL	International interest:	TITLE INSERT (INSULATOR), RECTANGULAR CONNECTOR, POLARIZED CENTER SCREW-LOCK, ELECTRICAL, 80 (SIZE 16 CRIMP REMOVABLE) PIN CONTACTS	MILITARY STANDARD MS 18163
Procurement Specification MIL-C-28748		SUPERSEDES:	SHEET 3 OF 3

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